REVCONNECT

Overview

RevConnect is a full-stack social networking platform designed to connect individuals, content creators, and businesses within a unified ecosystem. The application enables users to interact through posts, comments, likes, and connection requests while allowing creators and businesses to build an audience and promote their services.

The system is developed using **Spring Boot and Java**, following a structured **N-Layer architecture** to ensure maintainability, scalability, and separation of concerns. RevConnect integrates modern authentication mechanisms, optimized database design, and secure API communication to provide a robust social media platform.

The application supports three primary user roles:

- **Personal Users** – Users who interact socially by creating posts, connecting with others, and engaging with content.
- **Creators** – Users who publish content to grow followers and showcase their work.
- **Business Accounts** – Organizations that manage professional pages, publish promotional content, and interact with customers.

RevConnect demonstrates modern enterprise development practices including secure authentication, RESTful API communication, structured data modeling, and scalable backend architecture.

Technology Stack

The RevConnect platform is built using modern enterprise technologies:

Backend Framework

- Spring Boot 3

Programming Language

- Java 21

Templating Engine

- Thymeleaf (for server-side rendering)

Security

- Spring Security with JWT Authentication

Data Access

- Spring Data JPA with Hibernate ORM

Database

- Oracle SQL Database

Logging

- Log4j2

Build Tool

- Maven

Testing Framework

- JUnit and Mockito

This technology stack provides strong support for secure development, modular architecture, and scalable backend services.

Core Features

1. Authentication and Security

The platform provides secure user authentication and role-based access control.

Features include:

- Secure user registration and login
- Password encryption using BCrypt
- JWT-based authentication for secure API communication
- Role-based access control for Personal, Creator, and Business users
- Protection of sensitive endpoints through Spring Security filters

This ensures that only authorized users can access protected features of the system.

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2. Social Feed

RevConnect provides a dynamic social feed where users can view content shared by their connections and followed accounts.

Key capabilities include:

- Displaying posts from connected users
- Viewing personal posts on user profiles
- Supporting text-based posts with optional hashtags
- Filtering content based on user relationships

- Real-time update of feed content after new interactions

The feed mechanism improves user engagement and interaction within the platform.

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3. Social Interactions

The platform allows users to interact with content through multiple engagement features:

- **Like System** – Users can like or unlike posts.
- **Comment System** – Users can comment on posts and participate in discussions.
- **Reposts / Shares** – Content can be reshared with proper attribution.
- **Content Tracking** – Interactions are stored and linked to both users and posts.

These features encourage communication and content engagement.

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4. Networking Features

RevConnect includes networking functionality similar to professional social platforms.

Users can:

- Send connection requests
- Accept or reject connection requests
- View pending connection requests
- Follow or unfollow creators and business accounts
- View followers and following lists

This system allows users to build meaningful digital networks.

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5. Business and Creator Features

Creator and Business accounts receive enhanced capabilities.

Business accounts can:

- Create professional business pages
- Showcase products or services
- Share promotional posts
- Manage brand information such as address, contact information, and business hours

Creator accounts can:

- Build audiences through followers
- Publish specialized content

- Track engagement metrics

These features help businesses and creators expand their reach within the platform.

Security Implementation

RevConnect implements a comprehensive security model using **Spring Security**.

Security mechanisms include:

Authentication Strategy

- JWT tokens are generated after successful login.
- Each protected API request must include the token in the authorization header.
- Spring Security filters validate tokens before allowing access.

Identity Management

User authentication is managed through a custom user service that retrieves user details from the database and verifies credentials.

Password Protection

All user passwords are encrypted using **BCryptPasswordEncoder** before being stored in the Oracle database.

This security architecture protects sensitive user information and ensures secure API communication.

Data Handling and Quality

The application maintains a clean separation between internal data models and external API responses.

DTO Pattern

DTOs (Data Transfer Objects) are used to transfer data between layers without exposing internal entity structures.

Validation

Incoming requests are validated using standard validation annotations such as:

- @NotNull
- @Email
- @Size

This prevents invalid data from entering the system.

Testing Strategy

The project includes a structured testing approach:

- **Unit Testing** using JUnit for service layer validation
- **Mocking** with Mockito to isolate components
- **Integration Testing** to verify communication between application layers

This ensures code reliability and maintainability.

System Architecture

RevConnect follows a classic **N-Layer architecture** to ensure modular development.

The layers include:

Controller Layer

Handles incoming HTTP requests and exposes REST endpoints.

Service Layer

Implements the business logic and coordinates application workflows.

Repository Layer

Responsible for interacting with the database using Spring Data JPA.

Data Model Layer

Contains entity classes representing database tables and DTOs used for API communication.

This architecture ensures clear separation of responsibilities and improves system scalability.

Database Schema

The application uses **Oracle SQL Database** to store and manage all persistent data.

The schema is designed to support complex social networking relationships.

Core Entities

- Users
- Posts
- Business Pages
- Products

Relationship Entities

- Connections
- Followers
- Likes

- Comments
- Notifications

Foreign key constraints and normalized relationships ensure data consistency and efficient query performance.

Project Directory Structure

```
RevConnect_P2/
 ├– src/main/java/com/rev/app/
| ├– config/        # Application configuration
| ├– controller/     # Web controllers
| ├– rest/          # REST API controllers
| ├– service/        # Business logic
| ├– repository/     # Database access layer
| ├– entity/        # Database entities
| ├– dto/           # Data Transfer Objects
| ├– mapper/         # Entity to DTO mapping
| └ security/       # Security and JWT configuration
|
 ├– src/main/resources/
| ├– templates/      # Thymeleaf views
| ├– static/        # CSS, JavaScript, Images
| └ application.properties
|
└ pom.xml          # Maven configuration
```

REST API Overview

The system exposes REST APIs that allow communication between the frontend and backend.

Examples include:

Authentication APIs

POST /api/auth/authenticate

User APIs

GET /api/users/search
GET /api/users/me

Post APIs

GET /api/posts/feed
POST /api/posts
POST /api/posts/{id}/like

These APIs follow REST principles and return JSON responses.

Exception Handling

The system implements centralized exception handling using a **GlobalExceptionHandler**.

Custom exceptions include:

- **ResourceNotFoundException (404)** – When requested data does not exist
- **UserAlreadyExistsException (409)** – When duplicate user registration occurs
- **AccessDeniedException (403)** – When users attempt unauthorized actions

This improves system stability and user experience.